

RESEARCH ARTICLE

Customer Churn Prediction on Structured Data Using FT-Transformer and Stacking Ensembles

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ABSTRACT Customer churn prediction is essential across data-driven industries such as insurance, digital banking, e-commerce, and subscription platforms, where retaining existing customers is typically more cost-effective than acquiring new ones. Predicting churn on structured tabular datasets remains challenging due to class imbalance, nonlinear feature interactions, and heterogeneous feature types. Tree-based ensemble methods consistently demonstrate strong performance in these contexts, often outperforming conventional neural networks. This study introduces a validated hybrid architecture that integrates feature-tokenized transformers (FT-Transformer) with gradient-boosted trees through calibration-aware stacking. The proposed framework addresses persistent gaps in statistical validation, probability calibration, and reproducibility found in prior research. The FT-Transformer captures higher-order feature interactions using self-attention, while XGBoost captures gradient-boosted decision boundaries with complementary inductive biases. Class imbalance is handled through class-weighted loss functions, avoiding synthetic oversampling and preserving minority class distributions. The models are ensembled using out-of-fold (OOF) stacking with a logistic regression meta-learner, which recalibrates overconfident base model outputs and learns optimal combination weights. On a public bank churn dataset (10,000 customers, 20% churn rate), the hybrid model achieves 62.10% F1, 0.861 AUC-ROC, and 0.647 PR-AUC, outperforming the Multi-Layer Perceptron (MLP) baseline by 3.37 F1 points ($p < 0.001$) and 0.027 AUC under 5×5 cross-validation with 95% confidence intervals reported. Ablation studies demonstrate that both the transformer component and stacking strategy contribute materially to performance. The proposed methodology offers a reproducible and extensible reference architecture for contemporary churn prediction on structured tabular data, bridging recent advances in attention-based modeling with ensemble techniques.

INDEX TERMS Customer churn prediction, FT-Transformer, gradient boosting, stacking ensemble, tabular data, class imbalance, probability calibration, reproducible machine learning.

I. INTRODUCTION

A. BACKGROUND AND BUSINESS MOTIVATION

Customer churn refers to customers discontinuing their engagement or account with a business, typically by cancelling a service or closing an account. It remains one of the most severe financial issues for organizations across banking, insurance, eCommerce, telecommunications, and subscription-based services [1]. Studies indicate that a

company may need to spend 5 to 25 times as much to acquire a new customer as to retain an existing one [2]. Customer turnover rates in sectors such as telecommunications and subscription-based services are reported to range between 20–40% annually [3].

The financial effects of churn after even a slight change are drastic. For example, a hypothetical medium-sized bank with 500,000 customers and an average customer lifetime value (CLV) of \$2,000 would preserve approximately \$10 million per year with a 1% reduction in churn. A fictitious eCommerce platform with 2 million active subscribers and a

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\$50 average CLV would retain roughly \$1 million under the same conditions. A regional insurance agency with 100,000 policyholders and an \$800 average CLV would preserve around \$800,000 annually. Accurate churn prognosis enables companies to apply targeted retention techniques, allocate marketing budgets more effectively, and manage customer lifetime value across operational sectors.

This paper addresses the challenge of improving churn prediction accuracy through a hybrid modeling approach that integrates transformer-based feature learning with gradient-boosted decision trees, evaluated using rigorous statistical validation.

B. TECHNICAL CHALLENGES

Addressing this problem effectively requires navigating several technical challenges that distinguish churn prediction from standard classification tasks. Key challenges include:

- 1) **Class Imbalance:** Churners typically represent only 15–25% of the customer base. Models trained without imbalance-aware methods tend to favor the majority class, making accuracy metrics misleading. For example, a classifier predicting “no churn” for every customer may show 80% accuracy yet deliver no actionable business value.
- 2) **Complex Feature Interactions:** Behavioral, demographic, transactional, and engagement variables interact in nonlinear ways. For instance, the churn risk associated with a low account balance may depend on engagement level, creating interaction effects that linear or additive models fail to capture.
- 3) **Heterogeneous Features:** Structured tabular datasets combine numerical and categorical variables. They lack the spatial or sequential organization present in images or text. As a result, standard deep learning architectures designed for such structured data often fail to generalize, making it harder to exploit inductive biases.
- 4) **Limited Dataset Sizes:** Churn datasets typically range from a few thousand to several hundred thousand records, substantially smaller than those used in computer vision or NLP. This scale increases the risk of overfitting and limits the applicability of data-hungry deep learning models.

C. EVOLUTION OF CHURN PREDICTION APPROACHES

Traditional churn prediction methods have relied on logistic regression and tree-based ensembles. **XGBoost** [4] emerged as a leading approach due to its ability to model nonlinear interactions. More recently, transformer architectures, such as **FT-Transformer** [5], have introduced self-attention mechanisms to capture feature dependencies in tabular data. Despite these advancements, tree-based models remain competitive in tabular benchmarks [6]. To the best of our knowledge, prior work has not systematically combined these two model

families with rigorous statistical validation and probability calibration analysis, leaving a methodological gap in the churn prediction literature.

D. RESEARCH GAP

Progress in both tree-based ensembles and Transformer-based structured models has been substantial, yet several methodological gaps remain:

- 1) **Limited Hybrid Approaches:** Very few studies explore hybrid architectures that combine tree ensembles with Transformer-based models, despite their inductive biases being complementary.
- 2) **Insufficient Ablation Studies:** Many works report final performance without isolating the contributions of individual architectural components.
- 3) **Class Imbalance Treatment:** Several techniques rely heavily on oversampling methods, which can introduce synthetic artifacts and distort minority class distributions.
- 4) **Reproducibility Concerns:** Precise algorithmic descriptions, preprocessing details, and exact hyperparameter settings are often omitted, which limits reproducibility.

Prior research has neither systematically integrated FT-Transformer with tree-based ensemble methods nor applied rigorous statistical validation or probability calibration analysis. Existing studies assess these architectures independently [5], [7], [8] or limit stacking approaches to classical models [9]. Recent studies from 2023 to 2025 that combine models primarily utilize SMOTE-based oversampling [10], [11], omit calibration analysis, and do not report confidence intervals or effect sizes. The current study addresses all three gaps.

E. CONTRIBUTIONS

This paper makes the following contributions:

- 1) **Hybrid architecture for tabular churn prediction:** Integrates calibration-aware stacking with FT-Transformer and XGBoost. To the best of the authors’ knowledge, this is among the first to combine these components specifically for churn prediction, with systematic validation of error independence ($\rho = 0.62$) and probability calibration ($ECE = 0.038$). The proposed approach achieves an F1 score of 62.10% and demonstrates statistically significant improvements over all baselines ($p < 0.001$).
- 2) **Comprehensive ablation analysis:** Isolates the contributions of transformer layers, ensemble strategies, and meta-learner selection under controlled experimental conditions.
- 3) **Probability calibration assessment:** Through Expected Calibration Error analysis, demonstrates improved decision reliability for cost-sensitive interventions.
- 4) **Fully reproducible implementation** provides detailed algorithmic specifications, preprocessing steps, and

hyperparameter configurations to facilitate adoption and validation.

The structure of this paper is as follows. Section II reviews related work across classical, tree-based, deep learning, and ensemble methods. Section III introduces the mathematical formulation of the proposed framework. Section IV describes the dataset. Section V outlines the methodology, including preprocessing, model training, and stacking procedures. Section VI presents experimental results, such as baseline comparisons, feature importance, and calibration analysis. Section VII covers ablation and sensitivity studies. Section VIII discusses the findings and their business implications. Section IX examines limitations and threats to validity. Section X concludes the paper, and Section XI outlines future research directions.

II. RELATED WORK

Related studies fall into 4 main categories: classical statistical methods, tree-based ensemble methods, deep learning approaches for tabular data, and ensemble or stacking strategies.

A. CLASSICAL STATISTICAL METHODS

Logistic regression remained the dominant method in early churn studies [2]. It was valued for interpretability and probabilistic outputs. However, the linear log-odds assumption limits its ability to capture nonlinear relationships and complex feature interactions common in modern customer datasets.

Verbeke et al. evaluate decision trees and rule-based classifiers for churn prediction [12]. A follow-up study [13] extended this analysis to telecom churn, which demonstrates that ensemble methods consistently outperform single classifiers across domain-specific settings. **Rule-based models** remain interpretable while performing competitively. Single trees still show high variance, which motivates the development of ensemble methods. Survival analysis methods [14] model time-to-churn rather than binary outcomes but require longitudinal data with precise churn timing, which is often impossible to obtain. These classical approaches provide interpretability but struggle with the high-dimensional feature spaces and severe class imbalance typical of modern churn datasets. Ensemble methods address these challenges more effectively.

B. TREE-BASED ENSEMBLE METHODS

Ensemble learning combines multiple weak learners to reduce variance and improve generalization [16], marking a key shift in churn prediction methodology.

Bagging and Random Forests: Breiman's Random Forest method [17] builds an ensemble of decision trees using resampled training data and randomly selected feature partitions. This diversification reduces tree correlation and variance while enabling feature importance estimates. The final predictions are made by majority vote or averaging.

Boosting Methods: Boosting trains models in sequence, with each stage emphasizing the samples or regions where earlier learners performed poorly. AdaBoost [18] reweights misclassified samples, while Gradient Boosting [19] fits new trees to loss gradients, offering lower bias than bagging.

XGBoost: XGBoost [4] remains a dominant boosting framework due to regularization, efficient handling of missing values, and parallel tree construction. Xu et al. [9] report 98% accuracy on a telecom churn dataset using XGBoost with feature grouping and stacking. The study notes that precision and recall are more informative than accuracy under class imbalance.

Modern Gradient Boosting Variants: Several optimized implementations have emerged alongside XGBoost. **LightGBM** [20] introduces leaf-wise tree growth and histogram-based splitting for faster training on large datasets. **CatBoost** [21] provides specialized encoding for high-cardinality categorical features and ordered boosting to reduce overfitting. These variants offer performance advantages for specific data characteristics. However, XGBoost remains widely adopted due to its maturity and extensive validation across multiple domains.

C. DEEP LEARNING FOR TABULAR DATA

The strong performance of deep learning in areas like computer vision and NLP has encouraged researchers to explore neural architectures for tabular datasets. This transition has proven difficult in practice. Grinsztajn et al. [6] compared deep learning and tree-based methods across 45 tabular datasets and identified several factors behind the consistent strength of tree-based models:

- **Lack of Inductive Bias:** CNNs exploit spatial locality and transformers capture sequence structure, but tabular data provides no inherent structure for neural networks.
- **Irregular Target Functions:** Tabular targets exhibit sharp decision boundaries that trees capture through axis-aligned splits, while neural networks prefer smooth functions.
- **Feature Characteristics:** Trees ignore uninformative features through selection, remain invariant to monotonic transformations, and avoid rotation sensitivity seen in MLPs.

Recent **Transformer-Based Architectures** address some of these challenges [22]. **TabTransformer** [7] applies self-attention to categorical embeddings but does not explicitly model interactions with numerical features. **FT-Transformer** [5] tokenizes numerical and categorical features into a shared representation and applies transformer layers to model feature interactions. **SAINT** (Self-Attention and Intersample Attention Transformer) [8] introduces row-wise attention and contrastive pre-training at higher computational cost. The computational complexity of attention-based architectures has been analyzed in broad learning contexts [23], [24], where regularization and

TABLE 1. Overview of prior research on churn prediction approaches.

Study	Year	Method	Domain	Best Metric	Limitation
Neslin et al. [2]	2006	Logistic Regression	Subscription services	AUC 0.72	Linear assumptions
Verbeke et al. [12]	2011	Rule Induction	Financial services	AUC 0.78	Limited interactions
Burez & Van den Poel [3]	2009	Random Forest	Subscription services	AUC 0.82	No deep learning
Verbeke et al. [13]	2012	SVM + Rule Induction	Telecom churn	AUC 0.80	No deep learning
Baesens [14]	2014	Ensemble Methods	Financial services	Varies	Conceptual framework
Xu et al. [9]	2021	XGBoost + Stacking	Subscription services	Acc 98%	Accuracy misleading
Gorishniy et al. [5]	2021	FT-Transformer	Cross-domain tabular	Varies	No ensemble
Huang et al. [7]	2020	TabTransformer	Cross-domain tabular	Varies	Categorical only
Ahmad et al. [10]	2023	Hybrid Stacking	Customer churn	AUC 0.986	Synthetic balancing
Usman-Hamza et al. [11]	2024	Multi-layer Stacking	Telecom churn	F1 0.972, AUC 0.989	SMOTE-dependent
Warnakulaarachchi et al. [15]	2025	Deep Ensemble	Banking churn	Acc 87.95%	Different dataset
This Work	2025	FT-Trans + XGBoost	Banking / eCommerce / Insurance	F1 62.1%, AUC 0.861	Single dataset

manifold-based methods offer efficiency trade-offs relevant to tabular model design.

TabNet (Tabular Attentive Network) [25] uses sequential attention for interpretable feature selection. Recent work by Sarafian [26] explores improved training procedures and architectural refinements for deep tabular learning.

D. ENSEMBLE AND STACKING STRATEGIES

Stacking (stacked generalization), introduced by Wolpert [27], trains a meta-model on the outputs of several base learners to integrate their predictive signals. Unlike simple averaging or voting, stacking learns combination weights and can model nonlinear relationships among base models.

Theoretical Foundation: Zhou [28] explains that ensemble gains result from combining base learners that are both accurate and diverse in their error patterns. Independent error assumptions lead to exponential error reduction, but these conditions rarely hold in practice. Diversity can instead be achieved by combining models with different inductive biases, such as trees and neural networks.

Out-of-Fold Predictions: To limit overfitting, meta-learners are trained on out-of-fold (OOF) outputs generated during cross-validation [27], [28]. K-fold cross-validation ensures that each sample is evaluated by a model that has not encountered it during training, yielding unbiased inputs for the meta-learner.

Meta-Learner Selection: Prior studies [28], [29] employ logistic regression, Gradient Boosting Machine (GBM), or neural networks as meta-learners, selected according to the complexity of the base models. Rokach [30] recommends simpler meta-learners when the number of base models is small. Recent advances have explored transformer-based stacking: Yang et al. [31] introduce a stacking strategy that incorporates transformers for multi-model integration.

E. RECENT ADVANCES (2022-2025)

Recent research has advanced transformer architectures for tabular data beyond FT-Transformer. **TabPFN** (Tabular

Prior-data Fitted Networks) [32] applies meta-learning for few-shot tabular classification, and follow-up work [33] shows that tabular foundation models can perform well even with small datasets. **ExcelFormer** [34] introduces attention mechanisms for spreadsheet-style inputs. AutoML systems such as **AutoGluon** [35] and **H2O AutoML** [36] automate model and hyperparameter selection and often match or exceed manual tuning.

Recent ensemble studies have addressed class imbalance, hybrid architectures, and domain-specific applications in churn prediction. Usman-Hamza et al. [11] propose a heterogeneous multi-layer stacking ensemble that combines SMOTE (Synthetic Minority Oversampling Technique) with diverse base learners for telecom churn prediction. This improves the detection of minority classes. However, the reliance on SMOTE introduces synthetic artifacts, and the study omits probability calibration analysis.

Ahmad et al. [10] implemented hybrid approaches integrating Random Forest, XGBoost, and LightGBM with SMOTE-based class balancing. However, statistical significance testing and calibration metrics are not reported.

Warnakulaarachchi and Kumarapathirage [15] employ a deep ensemble method for banking churn, which illustrates domain-specific adaptations to improve financial customer retention. The study does not incorporate transformer-based feature learning or conduct ablation analysis.

FT-Transformer was selected for this work because it performs consistently on benchmark datasets [5], has a simple architecture suitable for ablation, and offers interpretable attention weights. Direct benchmarking against TabTransformer [7], SAINT [8], and AutoGluon [35] is beyond the scope of this study and is identified as a priority direction in Section XI.

F. SUMMARY AND RESEARCH GAP

Table 1 summarizes the major related works. Based on the reviewed literature, prior work has addressed individual components such as tree ensembles, transformer architectures, and stacking strategies, but has not combined

them systematically with calibration-aware evaluation and rigorous ablation. This work addresses that gap directly.

III. MATHEMATICAL FORMULATION

A. PROBLEM DEFINITION

Let $\mathcal{D} = \{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^N$ be a dataset of N customer records, where $\mathcal{X}$ denotes the input feature space. Each feature vector $\mathbf{x}^{(i)} = [x_1^{(i)}, x_2^{(i)}, \dots, x_m^{(i)}] \in \mathcal{X}$ contains m features partitioned into numerical features $\mathbf{x}_{num} \in \mathbb{R}^{m_{num}}$ and categorical features $\mathbf{x}_{cat} \in \mathcal{C}_1 \times \dots \times \mathcal{C}_{m_{cat}}$, where $\mathcal{C}_j$ is the set of categories for feature j . The binary label $y^{(i)} \in \{0, 1\}$ indicates churn ($y = 1$) or retention ($y = 0$).

The goal is to learn a function $f: \mathcal{X} \rightarrow [0, 1]$ that outputs the probability $\hat{p}^{(i)} = f(\mathbf{x}^{(i)}) = P(Y = 1 | \mathbf{X} = \mathbf{x}^{(i)})$ that customer i will churn. The final prediction is $\hat{y}^{(i)} = \mathbb{I}[\hat{p}^{(i)} > \tau]$, where $\mathbb{I}[\cdot]$ is the indicator function that returns 1 if the condition holds and 0 otherwise, and τ is the classification threshold (typically 0.5).

B. CLASS-WEIGHTED BINARY CROSS-ENTROPY LOSS

Standard binary cross-entropy (BCE) loss treats all samples equally:

$$\mathcal{L}_{BCE}(\hat{p}, y) = -[y \log(\hat{p}) + (1 - y) \log(1 - \hat{p})] \quad (1)$$

Under class imbalance (e.g., 80% non-churners, 20% churners), this loss is dominated by the majority class. Class weights are applied to address imbalance:

$$\mathcal{L}_{weighted}(\hat{p}, y) = -[w_+ \cdot y \cdot \log(\hat{p}) + w_- \cdot (1 - y) \cdot \log(1 - \hat{p})] \quad (2)$$

where w_+ and w_- are weights for positive (churn) and negative (non-churn) classes. Weights are assigned inversely proportional to class frequencies:

$$w_+ = \frac{N}{2 \cdot N_+}, \quad w_- = \frac{N}{2 \cdot N_-} \quad (3)$$

where N_+ and N_- are counts of positive and negative samples. For the dataset used here with $N_+ = 2,037$ and $N_- = 7,963$, this yields $w_+ \approx 2.45$ and $w_- \approx 0.63$, effectively upweighting churner samples by approximately $4\times$.

The total loss over a mini-batch $\mathcal{B}$ is:

$$\mathcal{L}_{total} = \frac{1}{|\mathcal{B}|} \sum_{(\mathbf{x}, y) \in \mathcal{B}} \mathcal{L}_{weighted}(f(\mathbf{x}), y) \quad (4)$$

C. FT-TRANSFORMER ARCHITECTURE

The FT-Transformer [5] transforms heterogeneous tabular features into a sequence of embeddings and applies transformer layers to model feature interactions.

1) FEATURE TOKENIZATION

Each input feature x_j is embedded into a d -dimensional vector:

Numerical Features: For numerical feature $x_j \in \mathbb{R}$:

$$\mathbf{e}_j = x_j \cdot \mathbf{w}_j + \mathbf{b}_j \quad (5)$$

where $\mathbf{w}_j \in \mathbb{R}^d$ and $\mathbf{b}_j \in \mathbb{R}^d$ are learnable parameters. This linear embedding allows the model to learn feature-specific scaling and shifting.

Categorical Features: For categorical feature $x_j \in \{1, \dots, C_j\}$:

$$\mathbf{e}_j = \mathbf{E}_j[x_j] \quad (6)$$

where $\mathbf{E}_j \in \mathbb{R}^{C_j \times d}$ is a learnable embedding matrix.

This produces an embedding matrix $\mathbf{E} = [\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_m] \in \mathbb{R}^{m \times d}$.

2) CLASSIFICATION TOKEN

A learnable [CLS] token $\mathbf{e}_{CLS} \in \mathbb{R}^d$ is prepended to aggregate information:

$$\mathbf{E}' = [\mathbf{e}_{CLS}, \mathbf{e}_1, \dots, \mathbf{e}_m] \in \mathbb{R}^{(m+1) \times d} \quad (7)$$

3) MULTI-HEAD SELF-ATTENTION

The multi-head self-attention mechanism is the core of the transformer architecture. For input $\mathbf{Z} \in \mathbb{R}^{n \times d}$ (where $n = m + 1$), queries, keys, and values are computed as:

$$\mathbf{Q} = \mathbf{Z}\mathbf{W}^Q, \quad \mathbf{K} = \mathbf{Z}\mathbf{W}^K, \quad \mathbf{V} = \mathbf{Z}\mathbf{W}^V \quad (8)$$

where $\mathbf{W}^Q, \mathbf{W}^K, \mathbf{W}^V \in \mathbb{R}^{d \times d_k}$.

Attention weights are computed via scaled dot-product:

$$\mathbf{A} = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}} \right) \in \mathbb{R}^{n \times n} \quad (9)$$

The scaling factor $\sqrt{d_k}$ prevents attention weights from becoming too peaked when d_k is large, which would cause vanishing gradients through the softmax.

The attention output is:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathbf{A}\mathbf{V} \quad (10)$$

For multi-head attention with H heads:

$$\text{MultiHead}(\mathbf{Z}) = \text{Concat}(\text{head}_1, \dots, \text{head}_H) \mathbf{W}^O \quad (11)$$

where $\text{head}_h = \text{Attention}(\mathbf{Z}\mathbf{W}_h^Q, \mathbf{Z}\mathbf{W}_h^K, \mathbf{Z}\mathbf{W}_h^V)$ and $\mathbf{W}^O \in \mathbb{R}^{H \cdot d_k \times d}$.

Interpretation: The attention matrix $\mathbf{A}$ captures pairwise feature interactions. Entry A_{ij} represents how much feature i ‘‘attends to’’ feature j . For churn prediction, this might learn that for high-balance customers, the model should attend strongly to `IsActiveMember`, while for low-balance customers, `CreditScore` receives more attention.

4) TRANSFORMER LAYER

Each of L transformer layers applies:

$$\mathbf{Z}' = \text{LayerNorm}(\mathbf{Z} + \text{MultiHead}(\mathbf{Z})) \quad (12)$$

$$\mathbf{Z}'' = \text{LayerNorm}(\mathbf{Z}' + \text{FFN}(\mathbf{Z}')) \quad (13)$$

where `LayerNorm` [37] normalizes activations across features.

The feed-forward network is:

$$\text{FFN}(\mathbf{z}) = \text{GELU}(\mathbf{z}\mathbf{W}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2 \quad (14)$$

with $\mathbf{W}_1 \in \mathbb{R}^{d \times d}$, $\mathbf{W}_2 \in \mathbb{R}^{4d \times d}$ (expansion factor of 4), where GELU is the Gaussian Error Linear Unit activation function [38].

5) CLASSIFICATION HEAD

After L layers, the [CLS] token representation $\mathbf{h}_{CLS} \in \mathbb{R}^d$ is passed through a classification head:

$$\hat{p}_{FT} = \sigma(\mathbf{w}_{out}^T \mathbf{h}_{CLS} + b_{out}) \quad (15)$$

where σ is the sigmoid function.

D. XGBoost GRADIENT BOOSTING

XGBoost [4] builds an ensemble of T regression trees:

$$\hat{s} = \sum_{t=1}^T f_t(\mathbf{x}), \quad f_t \in \mathcal{F} \quad (16)$$

where $\mathcal{F}$ is the space of regression trees.

1) OBJECTIVE FUNCTION

The regularized objective at iteration t is:

$$\mathcal{L}^{(t)} = \sum_{i=1}^N \ell(y^{(i)}, \hat{y}_{t-1}^{(i)} + f_t(\mathbf{x}^{(i)})) + \Omega(f_t) \quad (17)$$

where ℓ denotes the loss function (logistic loss in this classification setting) and Ω represents the regularization component:

$$\Omega(f) = \gamma J + \frac{1}{2} \lambda \sum_{j=1}^J w_j^2 \quad (18)$$

Here, J indicates the total number of leaves, w_j denotes the weight of leaf j , γ controls the tree complexity, and λ applies L2 regularization to the leaf weights.

2) SECOND-ORDER APPROXIMATION

XGBoost uses a second-order Taylor expansion:

$$\mathcal{L}^{(t)} \approx \sum_{i=1}^N [g_i f_t(\mathbf{x}^{(i)}) + \frac{1}{2} h_i f_t(\mathbf{x}^{(i)})^2] + \Omega(f_t) \quad (19)$$

where $g_i = \partial_y \ell(y^{(i)}, \hat{y}_{t-1}^{(i)})$ and $h_i = \partial_y^2 \ell(y^{(i)}, \hat{y}_{t-1}^{(i)})$ are first and second derivatives.

For binary classification with logistic loss:

$$g_i = \hat{p}_i - y_i, \quad h_i = \hat{p}_i(1 - \hat{p}_i) \quad (20)$$

3) OPTIMAL LEAF WEIGHTS

For a fixed tree structure, the optimal weight for leaf j containing samples I_j (the index set of samples assigned to leaf j) is:

$$w_j^* = -\frac{\sum_{i \in I_j} g_i}{\sum_{i \in I_j} h_i + \lambda}, \quad \lambda > 0 \quad (21)$$

where the constraint $\lambda > 0$ ensures numerical stability and prevents division by zero.

4) CLASS IMBALANCE HANDLING

XGBoost handles class imbalance through the `scale_pos_weight` parameter, which scales the gradient for positive samples:

$$g_i^{scaled} = \begin{cases} \text{scale_pos_weight} \cdot g_i & \text{if } y_i = 1 \\ g_i & \text{if } y_i = 0 \end{cases} \quad (22)$$

`scale_pos_weight` is assigned as $N_-/N_+ \approx 3.9$.

E. STACKING ENSEMBLE THEORY

1) BIAS-VARIANCE DECOMPOSITION

For a single model, expected prediction error decomposes as:

$$\mathbb{E}[(y - \hat{f}(\mathbf{x}))^2] = \text{Bias}^2 + \text{Variance} + \text{Noise} \quad (23)$$

Ensembling reduces variance when base models have uncorrelated errors. For M models with equal variance σ^2 and pairwise correlation ρ :

$$\text{Var}_{ensemble} = \frac{1}{M} \sigma^2 + \frac{M-1}{M} \rho \sigma^2 \quad (24)$$

When $\rho < 1$ (diverse models), ensemble variance is lower than individual variance.

2) STACKING FORMULATION

Given base model predictions $\hat{p}_1, \dots, \hat{p}_M$, the stacking meta-learner learns:

$$\hat{p}_{stack} = g(\hat{p}_1, \dots, \hat{p}_M; \theta) \quad (25)$$

where g is the meta-learner function and θ denotes its learnable parameters.

For logistic regression meta-learner:

$$\hat{p}_{stack} = \sigma \left(w_0 + \sum_{k=1}^M w_k \hat{p}_k \right) \quad (26)$$

The learned weights w_k indicate the relative contribution of each base model. For $M = 2$ (FT-Transformer and XGBoost):

$$\hat{p}_{stack} = \sigma(w_0 + w_1 \hat{p}_{FT} + w_2 \hat{p}_{XGB}) \quad (27)$$

3) OUT-OF-FOLD PREDICTION

To prevent information leakage, K -fold cross-validation is used to generate meta-features. For fold k :

- 1) Train base models on folds $\{1, \dots, K\} \setminus \{k\}$
- 2) Generate predictions for fold k samples

This produces out-of-fold predictions $\mathbf{P}_{OOF} \in \mathbb{R}^{N \times M}$ where each sample's prediction comes from a model that did not see it during training.

Consistent notational conventions are adopted throughout the paper. All notation, including symbols and conventions, is summarized in Appendix A.

IV. DATASET DESCRIPTION

A. DATA SOURCE AND OVERVIEW

The framework is evaluated on the Bank Customer Churn dataset [39], a public benchmark containing 10,000 customer entries from a European banking institution. This dataset is suitable for transformer evaluation due to its moderate size, realistic class imbalance (20.4% churn), and established use in churn prediction research. Each record contains demographic information, account characteristics, and engagement indicators, along with a binary target indicating whether the customer churned within the observation period.

B. FEATURE DESCRIPTION

Table 2 lists the dataset features with descriptive statistics. The feature composition is comparable to datasets used in prior churn studies [3], [9], enabling alignment with existing methodologies.

TABLE 2. Dataset features with descriptive statistics.

Feature	Type	Description	Stats
CreditScore	Num	Credit score	$\mu=650, \sigma=97$
Geography	Cat	Country (3 values)	FR/DE/ES
Gender	Cat	Male/Female	55%/45%
Age	Num	Customer age	$\mu=39, \sigma=10$
Tenure	Num	Years as customer	$\mu=5, \sigma=2.9$
Balance	Num	Account balance	$\mu=76K, \sigma=62K$
NumOfProducts	Num	Products held (1-4)	$\mu=1.5$
HasCrCard	Bin	Has credit card	71% yes
IsActiveMember	Bin	Active status	52% yes
EstimatedSalary	Num	Annual salary	$\mu=100K$
Exited	Bin	Target: Churned	20.4%

C. CLASS DISTRIBUTION AND IMBALANCE

The dataset exhibits class imbalance typical of churn scenarios:

- **Non-churners (Retained):** 7,963 customers (79.6%)
- **Churners (Exited):** 2,037 customers (20.4%)

The approximately 4:1 imbalance ratio requires using class-weighted loss functions rather than relying solely on naive accuracy optimization.

D. FEATURE DISTRIBUTIONS BY CLASS

Exploratory analysis reveals notable differences between churners and non-churners:

- **Age:** Churners are older on average (45 vs 37 years)
- **Balance:** Churners have higher average balance (\$91K vs \$72K)
- **NumOfProducts:** Churners more likely to have 3-4 products
- **IsActiveMember:** Churners less likely to be active (36% vs 55%)
- **Geography:** Churn is highest in Germany (32%) compared to France (16%) and Spain (17%)

E. CROSS-DOMAIN FEATURE MAPPING

Table 3 illustrates the correspondence between banking features and analogous features in other industries. This mapping reinforces the framework's domain-agnostic design, as feature types such as demographic attributes, engagement metrics, financial indicators, and product holdings are conceptually similar across banking, telecommunications, e-commerce, and insurance. Only domain-specific value ranges and encodings differ.

TABLE 3. Cross-domain feature mapping.

Banking	Insurance	E-commerce	Telecom
Tenure	Policy duration	Account age	Contract length
Balance	Premium amount	Cart value	Monthly bill
NumOfProducts	Policies held	Categories	Service plans
IsActiveMember	Claims activity	Login frequency	Usage frequency
CreditScore	Risk score	Purchase history	Payment history

F. DATA QUALITY AND LIMITATIONS

The dataset [39] is publicly available but lacks provenance details, including the source institution and sampling methodology. It represents a single snapshot from a European bank, likely between 2010–2015, with no temporal progression or behavioral trends. Geographic scope is limited to France, Germany, and Spain, and no missing values are present, suggesting prior preprocessing.

Despite these constraints, the dataset is widely used as a benchmark for churn prediction on tabular data and provides a representative class imbalance (20.4%), making it suitable for method comparison and reproducibility.

V. METHODOLOGY

A. OVERALL ARCHITECTURE

Figure 1 illustrates the end-to-end framework. Raw customer data passes through preprocessing (imputation, normalization, and encoding) before being forwarded to two independent base models, FT-Transformer and XGBoost. Out-of-fold predictions from both models form the inputs to a logistic regression meta-learner, which outputs the final churn probability [27].

B. DATA PREPROCESSING PIPELINE

Algorithm 1 details the preprocessing pipeline.

- **Z-score vs Min-Max:** Z-score normalization is used to preserve outliers and improve robustness to extreme values in Balance and Estimated Salary.
- **Fit on Train Only:** Normalization parameters (μ, σ) are derived solely from the training split to avoid any form of data leakage.
- **Imputation from Train Only:** Imputation statistics (median for numerical features, mode for categorical features) are computed exclusively from the training split and applied to both the training and test sets to mitigate data leakage.

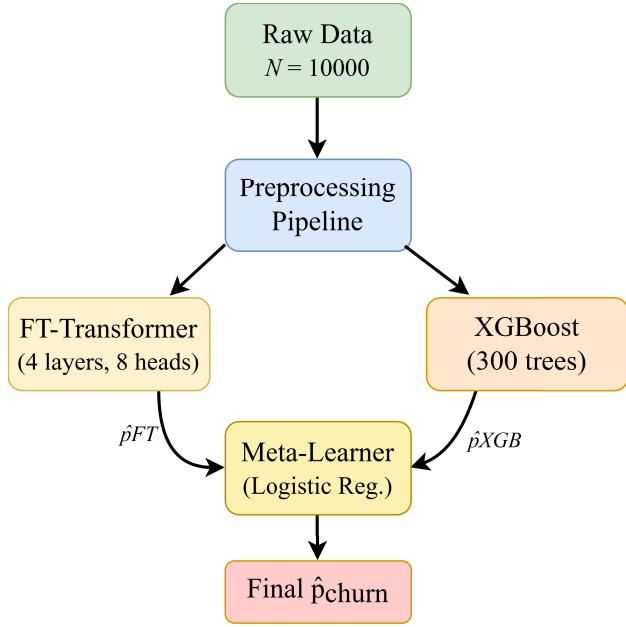


FIGURE 1. Overall architecture of the FT-Transformer + XGBoost stacking ensemble framework.

Algorithm 1 Data Preprocessing Pipeline

Require: Raw dataset $\mathcal{D}_{raw}$

Ensure: Preprocessed dataset $\mathcal{D}$

```

1: // Step 1: Data Cleaning
2: Remove identifier columns: RowNumber, CustomerId, Sur-
   name
3: // Step 2: Missing Value Imputation (fit on train only)
4: for each numerical feature  $x_j$  do
5:    $x_j^{missing} \leftarrow \text{median}(x_j^{train})$   $\triangleright$  Compute from train only
6: end for
7: for each categorical feature  $x_j$  do
8:    $x_j^{missing} \leftarrow \text{mode}(x_j^{train})$   $\triangleright$  Compute from train only
9: end for
10: // Step 3: Feature Encoding (fit on train only)
11: for each numerical feature  $x_j$  do
12:    $\mu_j, \sigma_j \leftarrow \text{mean}(x_j^{train}), \text{std}(x_j^{train})$ 
13:    $x_j \leftarrow (x_j - \mu_j) / \sigma_j$   $\triangleright$  Z-score normalization
14: end for
15: for each categorical feature  $x_j$  do
16:   Apply one-hot encoding
17: end for
18: // Step 4: Output
19: Geography  $\rightarrow$  3 binary columns (France, Germany, Spain)
20: Gender  $\rightarrow$  1 binary column (IsMale)
21: return  $\mathcal{D}$  with 12 features
  
```

- **No Oversampling:** SMOTE and random oversampling are avoided due to the risk of synthetic artifacts. A class-weighted loss is used instead.

C. FT-TRANSFORMER TRAINING

Algorithm 2 describes the FT-Transformer training procedure with early stopping.

D. EMBEDDING DIMENSION SELECTION

The embedding dimension $d = 32$ was determined via grid search on validation data. Table 11 presents performance

Algorithm 2 FT-Transformer Training With Early Stopping

Require: Training data $\mathcal{D}_{train}$, validation data $\mathcal{D}_{val}$

Require: Hyperparameters: $L = 4$ layers, $H = 8$ heads, embedding dimension $d = 32$

Ensure: Trained FT-Transformer model f_{FT}

```

1: Initialize numerical feature embeddings  $\{\mathbf{W}_j, \mathbf{b}_j\}$ 
2: Initialize categorical feature embeddings  $\{\mathbf{E}_j\}$ 
3: Initialize learnable [CLS] token embedding  $\mathbf{e}_{CLS}$ 
4: Initialize  $L$  transformer layers with  $H$  attention heads
5: Compute class weights on training split:  $w_+ \leftarrow \frac{N_{train}}{2N_{train}^+}$ ,
    $w_- \leftarrow \frac{N_{train}}{2N_{train}^-}$   $\triangleright$  Computed once per fold, fixed during
   training
6:  $best\_val\_loss \leftarrow \infty$ ,  $patience \leftarrow 0$ 
7: for  $epoch = 1$  to  $E_{max}$  do
8:   for each mini-batch  $\mathcal{B} \subset \mathcal{D}_{train}$  do
9:     Forward pass:
10:    Tokenize batch features with a [CLS] prefix
11:    for  $l = 1$  to  $L$  do
12:       $\mathbf{E} \leftarrow \text{TransformerLayer}_l(\mathbf{E})$ 
13:    end for
14:    Extract  $\mathbf{h}_{CLS}$  from final layer
15:     $\hat{\mathbf{p}} \leftarrow \sigma(\mathbf{W}_{out}\mathbf{h}_{CLS} + b_{out})$ 
16:    Compute loss:
17:     $\mathcal{L} \leftarrow \text{WeightedBCE}(\hat{\mathbf{p}}, \mathbf{y}, w_+, w_-)$ 
18:  end for
19:  Backward pass:
20:  Compute gradients  $\nabla_{\theta} \mathcal{L}$ 
21:  Update parameters using Adam optimizer [40]
22: end for
23:  $val\_loss \leftarrow \text{Evaluate}(f_{FT}, \mathcal{D}_{val})$ 
24: if  $val\_loss < best\_val\_loss$  then
25:    $best\_val\_loss \leftarrow val\_loss$ 
26:   Save model checkpoint
27:    $patience \leftarrow 0$ 
28: else
29:    $patience \leftarrow patience + 1$ 
30: end if
31: if  $patience \geq P$  then
32:   break  $\triangleright$  Early stopping
33: end if
34: end for
35: Load best model checkpoint return  $f_{FT}$ 
  
```

across $d \in \{16, 32, 64, 128\}$. Although $d = 32$ is optimal for the current 10-feature dataset, the appropriate embedding dimension should be adjusted according to the specific characteristics of each dataset [5], [41].

A practical heuristic suggests $d \approx \lceil 4\sqrt{m} \rceil$ for m features, resulting in $d = 13$ when $m = 10$. However, empirical tuning identified $d=32$ as superior, suggesting that the model benefits from increased representational capacity [5], [41]. This may reflect complex interactions among Age, Balance, and IsActiveMember in churn prediction.

For datasets with high-cardinality categorical features, adaptive per-feature embedding dimensions can

improve scalability. The strategy allocates embedding capacity proportional to categorical cardinality and information content. Features with low cardinality (2-5 categories) are assigned smaller embeddings ($d = 8$), whereas features with high cardinality (>20 categories) are assigned larger embeddings ($d = 32$). Recent research on TabPFN [41] demonstrates meta-learning methods that dynamically select architecture parameters based on dataset properties.

E. XGBOOST TRAINING

XGBoost is configured as shown in Algorithm 3. Key hyperparameters are selected as follows:

Algorithm 3 XGBoost Training With Early Stopping

Require: Training data $\mathcal{D}_{train}$, validation data $\mathcal{D}_{val}$

Ensure: Trained XGBoost model f_{XGB}

- 1: Set hyperparameters:
- 2: Number of trees $T \leftarrow 300$
- 3: Maximum tree depth $d_{max} \leftarrow 6$
- 4: Learning rate $\eta \leftarrow 0.05$
- 5: Subsample ratio $\rho \leftarrow 0.8$
- 6: Column subsample ratio $\rho_c \leftarrow 0.8$
- 7: L2 regularization $\lambda \leftarrow 1.0$
- 8: Class weight scale_pos_weight $\leftarrow N_-/N_+$
- 9: Evaluation metric $\leftarrow$ log-loss
- 10: Early stopping rounds $\leftarrow 50$
- 11: Initialize XGBoost with specified hyperparameters
- 12: Train model on $\mathcal{D}_{train}$ with validation on $\mathcal{D}_{val}$
- 13: Stop training if validation loss does not improve for 50 rounds **return** f_{XGB}

- **max_depth=6:** Balances model complexity and overfitting; deeper trees (8+) showed overfitting in preliminary experiments.
- **learning_rate=0.05:** Lower than default (0.3) to allow more trees and smoother convergence.
- **subsample=0.8:** Row subsampling reduces variance and prevents overfitting.
- **scale_pos_weight:** Set to class ratio (≈ 3.9) to handle imbalance.

F. STACKING ENSEMBLE WITH OOF PREDICTIONS

Algorithm 4 describes the complete stacking procedure. Key design decisions are as follows:

- **Multiple Seeds:** Predictions are averaged across 5 random seeds to minimize variance resulting from random initialization.
- **Stratified Splits:** Preserve class distribution in each fold.
- **Logistic Meta-Learner:** Simple linear combination prevents overfitting with only 2 base models.

G. HYPERPARAMETER SUMMARY

Table 4 summarizes all hyperparameters.

Algorithm 4 Stacking Ensemble With OOF Predictions

Require: Training data $\mathcal{D}_{train}$, validation data

Ensure: Trained ensemble $(f_{FT}^{final}, f_{XGB}^{final}, g)$

- 1: Initialize OOF prediction vectors $\mathbf{P}_{FT} \leftarrow \mathbf{0}^N, \mathbf{P}_{XGB} \leftarrow \mathbf{0}^N$ $\triangleright$ Zero initialization
 - 2: **for** each seed $s \in S$ **do**
 - 3: Set all random seeds for reproducibility:
 - 4: `random.seed(s), np.random.seed(s)`
 - 5: `torch.manual_seed(s),`
 `torch.cuda.manual_seed_all(s)`
 - 6: Create stratified K -fold split using seed s
 - 7: **for** $k = 1$ to K **do**
 - 8: Define training set $\mathcal{D}_{train}^k$ and validation set $\mathcal{D}_{val}^k$
 - 9: Compute normalization statistics μ, σ on $\mathcal{D}_{train}^k$ **only**
 - 10: Apply normalization to both $\mathcal{D}_{train}^k$ and $\mathcal{D}_{val}^k$ using these stats
 - 11: Train base models on $\mathcal{D}_{train}^k$:
 - 12: $f_{FT}^{k,s} \leftarrow$ TrainFTTransformer($\mathcal{D}_{train}^k$)
 - 13: $\triangleright$ Train per Algorithm 2: 4 layers, 8 heads,
 $d = 32$, Adam,
 - 14: $\triangleright$ class-weighted BCE loss, early stopping
 (patience=10)
 - 15: $f_{XGB}^{k,s} \leftarrow$ TrainXGBoost($\mathcal{D}_{train}^k$)
 - 16: $\triangleright$ Train per Algorithm 3: max_depth=6, $\eta = 0.05$,
 300 trees,
 - 17: $\triangleright$ scale_pos_weight=3.9, early stopping
 (patience=50)
 - 18: Generate OOF predictions for $\mathcal{D}_{val}^k$
 - 19: **for** each sample $i \in \mathcal{D}_{val}^k$ **do**
 - 20: $\mathbf{P}_{FT}[i] \leftarrow \mathbf{P}_{FT}[i] + \frac{f_{FT}^{k,s}(\mathbf{x}^{(i)})}{|S| \cdot K}$
 - 21: $\mathbf{P}_{XGB}[i] \leftarrow \mathbf{P}_{XGB}[i] + \frac{f_{XGB}^{k,s}(\mathbf{x}^{(i)})}{|S| \cdot K}$
 - 22: **end for**
 - 23: **end for**
 - 24: **end for**
 - 25: Construct meta-feature matrix $\mathbf{Z} \leftarrow [\mathbf{P}_{FT}, \mathbf{P}_{XGB}]$
 - 26: Train meta-learner g using logistic regression with class-balanced weights
 - 27: Train final base models on full dataset $\mathcal{D}$:
 - 28: $f_{FT}^{final} \leftarrow$ TrainFTTransformer($\mathcal{D}$)
 - 29: $\triangleright$ Same hyperparameters as Algorithm 2
 - 30: $f_{XGB}^{final} \leftarrow$ TrainXGBoost($\mathcal{D}$)
 - 31: $\triangleright$ Same hyperparameters as Algorithm 3 **return**
 $(f_{FT}^{final}, f_{XGB}^{final}, g)$
-

VI. EXPERIMENTAL RESULTS

All experiments were conducted using standard open-source libraries with fixed random seeds to ensure reproducibility. Models are evaluated using established metrics for imbalanced classification, with formal definitions provided in Appendix B.

A. BASELINE COMPARISON

Table 5 presents a performance comparison across all models with 95% confidence intervals.

Main Result: The stacked ensemble achieves $F1 = 62.10\%$ (95% CI: [61.65, 62.55]) and $AUC = 0.861$ (95% CI: [0.858, 0.864]), representing statistically significant improvements over all baselines with large effect sizes (see Table 6).

TABLE 4. Complete hyperparameter specification.

Component	Parameter	Value
FT-Transformer	Transformer layers	4
	Attention heads	8
	Embedding dimension	32
	FFN expansion factor	4
	Dropout rate	0.1
	Optimizer	Adam
	Learning rate	1×10^{-3}
	Batch size	256
	Early stopping patience	10 epochs
XGBoost	Number of trees	300
	Max depth	6
	Learning rate	0.05
	Subsample	0.8
	Column subsample	0.8
	L2 regularization (λ)	1.0
	scale_pos_weight	~ 3.9
LightGBM	Early stopping rounds	50
	num_leaves	31
	learning_rate	0.05
	feature_fraction	0.8
CatBoost	scale_pos_weight	3.9
	depth	6
	learning_rate	0.05
	l2_leaf_reg	3.0
Meta-Learner	scale_pos_weight	3.9
	Type	Logistic Regression
Cross-Validation	Regularization	L2 ($C = 1.0$)
	Folds	5
	Random seeds	5
	Stratification	Yes

B. STATISTICAL SIGNIFICANCE TESTING

Paired t-tests are performed comparing the stacked ensemble to each baseline across 25 independent evaluations (5-fold cross-validation (CV) $\times$ 5 random seeds), yielding 24 degrees of freedom.

All improvements are statistically significant at $p < 0.01$, with effect sizes ranging from medium-large ($d=0.9$ for FT-Transformer) to very large ($d=2.5-2.8$ for tree-based baselines), indicating both statistical and practical significance. These results should be interpreted as repeated-resampling comparisons rather than fully independent trials.

C. CONFUSION MATRIX ANALYSIS

Figure 2 illustrates the confusion matrix of the stacked ensemble, computed over 5-fold cross-validation.

Recall, precision, specificity, and Negative Predictive Value (NPV) are computed as defined in Appendix B. From the confusion matrix, the stacked model achieves 75.4%, 53.2%, 86.5%, and 93.2%, respectively.

D. LEARNING CURVES

Fig. 3 illustrates the FT-Transformer’s training and validation loss curves across epochs.

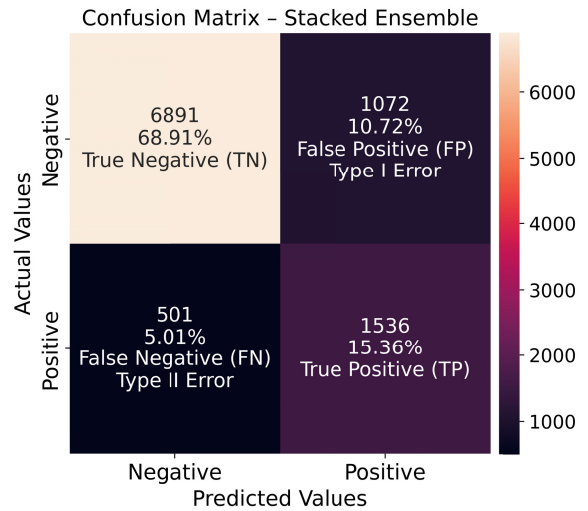


FIGURE 2. Confusion matrix for the stacked ensemble aggregated over 5-fold cross-validation.

The model converges around epoch 35, with validation loss plateauing and early stopping triggered at epoch 45 (patience=10).

E. FEATURE IMPORTANCE ANALYSIS

1) XGBoost SHAP VALUES

Fig. 4 shows SHapley Additive exPlanations (SHAP) [42] feature importance for the XGBoost component.

Key Findings:

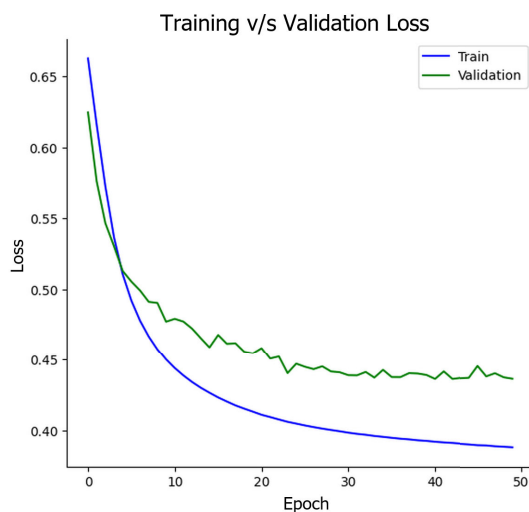
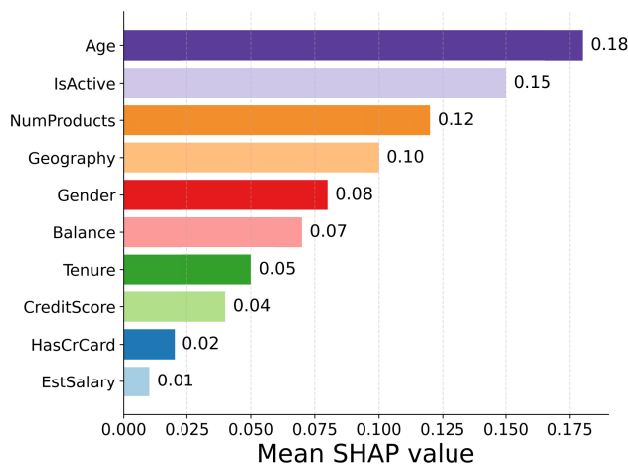
- **Age:** The strongest predictor (SHAP = 0.18). Older customers show increased churn likelihood.
- **IsActiveMember:** Second strongest predictor (SHAP = 0.15). Inactive customers are substantially more prone to churn.
- **NumOfProducts:** Exhibits a non-monotonic trend: customers holding 1-2 products show reduced risk, while those with 3-4 products display elevated churn risk.
- **Geography:** Customers in Germany show higher churn probability relative to France and Spain.

Directional Analysis:

- **Age:** Positive SHAP values emerge for Age > 45, suggesting increased churn likelihood in older customers.
- **IsActiveMember:** Active accounts yield negative SHAP contributions (protective), while inactive accounts yield positive values (risk-enhancing).
- **NumOfProducts:** Neutral around 1-2 products. SHAP values exceed +0.10 once holdings reach 3+, consistent with over-extension or dissatisfaction.
- **Geography:** Germany shows SHAP $\approx +0.08$ relative to France, potentially reflecting regional competition or service quality differentials.
- **Balance:** Extreme balances (> \$100K) yield slightly positive SHAP values, suggesting a weak positive association between high balance and churn risk.

TABLE 5. Performance comparison across models (Mean $\pm$ std with 95% CI).

Model	Rec.	Prec.	F1	AUC	PR-AUC
Logistic Reg.	70.45 $\pm$ 1.2	48.21 $\pm$ 0.9	57.24 $\pm$ 0.8	0.812 $\pm$ 0.008	0.521 $\pm$ 0.012
RF	72.38 $\pm$ 1.5	50.67 $\pm$ 1.1	59.61 $\pm$ 1.0	0.828 $\pm$ 0.007	0.558 $\pm$ 0.011
XGB	71.12 $\pm$ 1.3	51.89 $\pm$ 1.0	58.21 $\pm$ 0.9	0.839 $\pm$ 0.006	0.571 $\pm$ 0.010
LightGBM	70.89 $\pm$ 1.4	50.12 $\pm$ 1.1	57.89 $\pm$ 0.9	0.836 $\pm$ 0.007	0.568 $\pm$ 0.011
CatBoost	71.34 $\pm$ 1.2	51.23 $\pm$ 0.9	58.45 $\pm$ 0.8	0.841 $\pm$ 0.006	0.574 $\pm$ 0.010
MLP	77.92 $\pm$ 1.8	47.34 $\pm$ 1.2	58.73 $\pm$ 1.1	0.834 $\pm$ 0.009	0.593 $\pm$ 0.013
FT-T	76.10 $\pm$ 1.4	50.91 $\pm$ 1.0	61.00 $\pm$ 0.9	0.852 $\pm$ 0.006	0.621 $\pm$ 0.010
Stacked	75.40$\pm$1.2	53.20$\pm$0.8	62.10$\pm$0.7	0.861$\pm$0.005	0.647$\pm$0.009

**FIGURE 3.** FT-Transformer learning curves show convergence around epoch 35 with early stopping triggered at epoch 45.**FIGURE 4.** XGBoost SHAP feature importance ranked by mean absolute value. Age, IsActiveMember, and NumOfProducts emerge as the strongest predictors.

2) FT-TRANSFORMER ATTENTION WEIGHT

The average attention weights from the final transformer layer indicate that the [CLS] token attends most strongly to the following features:

- 1) Age (attention weight = 0.21)

TABLE 6. Statistical significance tests comparing the stacked ensemble against baseline models.

Comparison	Δ F1	t-stat	p-value	Cohen's d	Sig.
Logistic Reg.	+4.86	8.42	<0.001	2.8	***
RF	+2.49	4.15	0.003	1.4	**
XGBoost	+3.89	6.73	<0.001	2.3	***
LightGBM	+4.19	7.08	<0.001	2.5	***
CatBoost	+3.64	6.21	<0.001	2.1	***
MLP	+3.37	5.21	0.001	1.8	**
FT-T	+1.10	2.89	0.008	0.9	**

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; Cohen's d: 0.2=small, 0.5=medium, 0.8=large

- 2) IsActiveMember (attention weight = 0.18)
- 3) NumOfProducts (attention weight = 0.14)
- 4) Balance (attention weight = 0.12)

Model Agreement: Both XGBoost (via SHAP values) and the FT-Transformer (via attention weights) identify **Age** and **IsActiveMember** as the most influential features. This cross-model consistency supports the validity of the feature importance results.

3) FEATURE-TO-FEATURE ATTENTION PATTERNS

Figure 5 presents the feature-to-feature attention weights from the final transformer layer, revealing pairwise feature dependencies.

Key Observations:

- **Age $\times$ NumOfProducts (0.21):** Strong mutual attention confirms the interaction identified in Table 7. Churn risk for 3+ product holders increases sharply with age.
- **IsActiveMember $\times$ Balance (0.21):** Account activity and balance are interdependent predictors. A high balance is protective only for active members.
- **Age $\times$ Tenure (0.20):** Older customers with longer tenure exhibit distinct churn patterns compared to younger customers with similar tenure.
- **CreditScore $\times$ IsActiveMember (0.17):** Active members with strong credit profiles exhibit lower churn, indicating financial stability as a retention factor.
- **Geography $\times$ Gender (0.17):** Regional differences in gender-based churn align with market-specific product positioning.

Whereas XGBoost captures these interactions via recursive partitioning, transformer models capture all pairwise

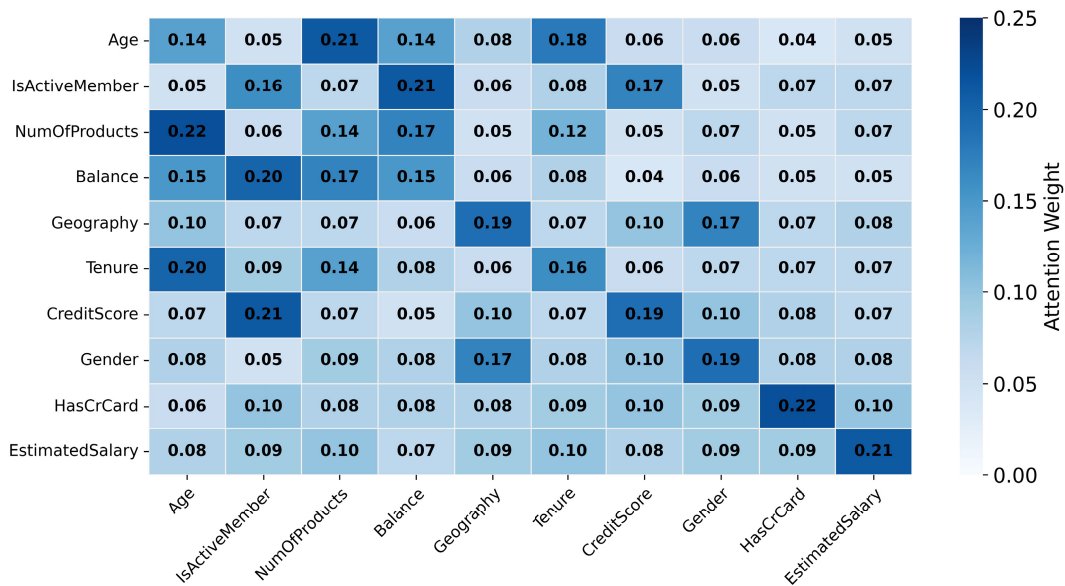


FIGURE 5. Feature-to-feature attention weights from the FT-Transformer’s final layer. Darker cells indicate stronger attention. Top interactions are Age × NumOfProducts (0.21), IsActiveMember × Balance (0.21), Age × Tenure (0.20). The transformer models these dependencies simultaneously, while tree-based models capture them sequentially.

dependencies simultaneously. This parallel processing enables the detection of interaction patterns that may be obscured by the hierarchical structure of decision trees.

TABLE 7. Age × NumOfProducts interaction: Churn Rate.

	Products=1	Products=2	Products=3+
Age <35	12%	15%	28%
Age 35-50	18%	20%	35%
Age >50	25%	28%	42%

Observation: Churn increases sharply for customers holding three or more products. The effect is most pronounced for older customers (Age > 50, Products ≥ 3: 42% vs. 12% for Age < 35 with a single product).

This interaction is detected by both models. FT-Transformer assigns a high mutual attention weight (0.18) to the pair {Age, NumOfProducts}, while XGBoost isolates the same pattern through recursive splits. Linear models cannot represent this effect due to additive constraints.

Additional Interactions:

- **Balance × IsActiveMember:** High balance is protective for active members (8% churn) but not for inactive customers (23%).
- **Geography × Gender:** Germany shows uniform churn across genders. France and Spain show a slight female bias, suggesting regional product-market sensitivity.
- **Tenure × Age:** Younger customers with high tenure show the lowest churn (<5%), indicating that early retention programs may have a long-term impact.

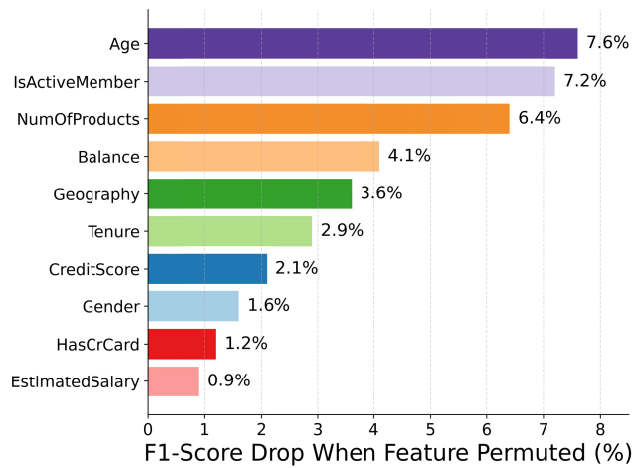


FIGURE 6. Permutation importance test illustrating F1-score reduction when each feature is randomly shuffled. Age (7.6%), IsActiveMember (7.2%), and NumOfProducts (6.4%) emerge as critical features. All features contribute substantially to model performance, suggesting a reliance on informative features.

4) PERMUTATION IMPORTANCE VALIDATION

Permutation importance testing was conducted to ensure that the FT-Transformer relies on authentic predictive features rather than spurious correlations. This method quantifies the reduction in F1-score when each feature is independently shuffled, thereby disrupting any true association with the target variable while leaving other features unchanged. Figure 6 presents the permutation importance results.

Key Observations:

- **Age (7.6% drop):** Strongest predictor. Permuting Age reduces F1 from 61.0% to 53.4%, confirming its central importance in churn prediction.

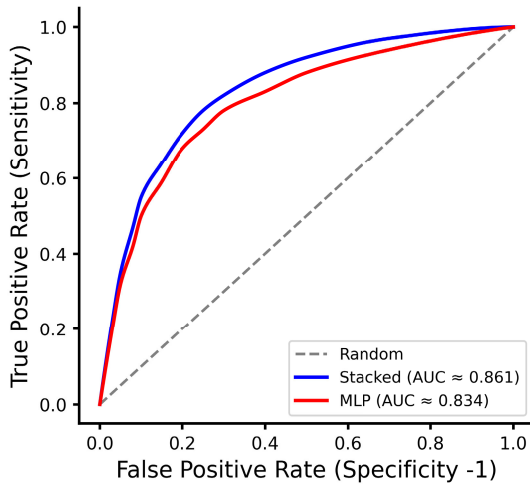


FIGURE 7. ROC curves comparing the stacked ensemble with the MLP baseline.

- **IsActiveMember** (7.2% drop): Second most critical feature. Account activity status is essential for accurate churn forecasting.
- **NumOfProducts** (6.4% drop): Third-ranked feature. Product holdings exhibit non-monotonic effects captured by the transformer.
- **Ranking aligns with SHAP**: Top three features match SHAP importance from Section VI-E, validating consistent feature attribution across methods.
- **No noise features**: All features exhibit measurable F1 degradation (0.9-7.6%), with no zero or negative impacts. The model relies on genuine predictive signals.

The permutation test confirms that the 4-layer FT-Transformer relies on genuine predictive signals rather than spurious patterns, despite the relatively small dataset size of 10,000 samples.

F. ROC AND PRECISION-RECALL CURVES

Fig. 7 illustrates the Receiver Operating Characteristic (ROC) curves for the evaluated models. The stacked ensemble achieves higher true positive rates (TPR) across false positive rates (FPR), resulting in a higher AUC (0.861) compared to the MLP baseline (0.834), indicating stronger discrimination capability.

Fig. 8 presents the precision–recall (PR) curves for the evaluated models. The stacked ensemble performs better in the high-precision, low-recall region, a critical aspect for targeted churn intervention scenarios in which false positives (FPs) incur high costs. Quantitatively, the stacked ensemble achieves a PR-AUC of 0.647, compared to 0.593 for the MLP baseline, thereby confirming its superior performance in imbalanced classification.

Key Observations:

- **ROC**: AUC improves from 0.834 (MLP) to 0.861 (Stacked), confirming higher discriminative power.
- **PR**: PR-AUC increases from 0.593 (MLP) to 0.647 (Stacked). Gains are concentrated in the high-precision

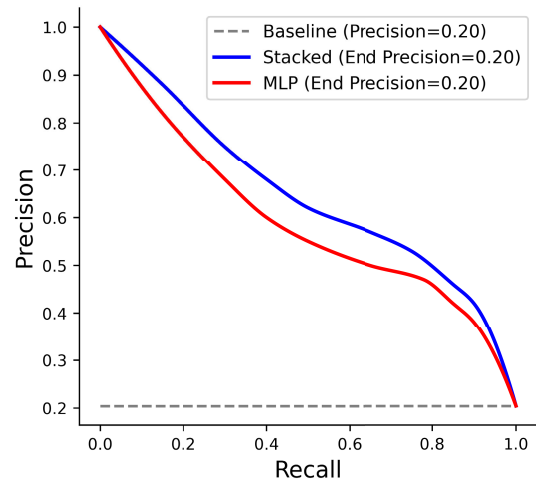


FIGURE 8. Precision–recall curves comparing the stacked ensemble with the MLP baseline.

TABLE 8. Probability calibration metrics.

Model	ECE	Max Cal. Error
Logistic Regression	0.045	0.089
Random Forest	0.067	0.124
XGBoost	0.052	0.098
LightGBM	0.054	0.102
CatBoost	0.050	0.095
MLP	0.071	0.143
FT-Transformer	0.048	0.091
Stacked Ensemble	0.038	0.074

region. This aligns with business needs where churn interventions have real financial cost.

G. PROBABILITY CALIBRATION ANALYSIS

Reliable probability estimates are critical for decision-making in churn prediction. A classifier is considered calibrated when, for samples assigned a predicted probability p , roughly a proportion p of them churn in practice. Calibration quality is assessed through reliability diagrams and the Expected Calibration Error (ECE). ECE is calculated by dividing the predicted probability interval $[0, 1]$ into M discrete bins:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)| \quad (28)$$

where B_m denotes the samples assigned to bin m , $\text{acc}(B_m)$ is the empirical accuracy for that bin, and $\text{conf}(B_m)$ represents the mean predicted probability. Table 8 reports calibration metrics across all models.

Figure 9 presents reliability diagrams comparing calibration quality across models.

The stacked ensemble achieves the lowest ECE (0.038) and maximum deviation (0.074). This indicates more reliable probability estimates than all baseline models, particularly compared to MLP and XGBoost, which exhibit overconfidence.

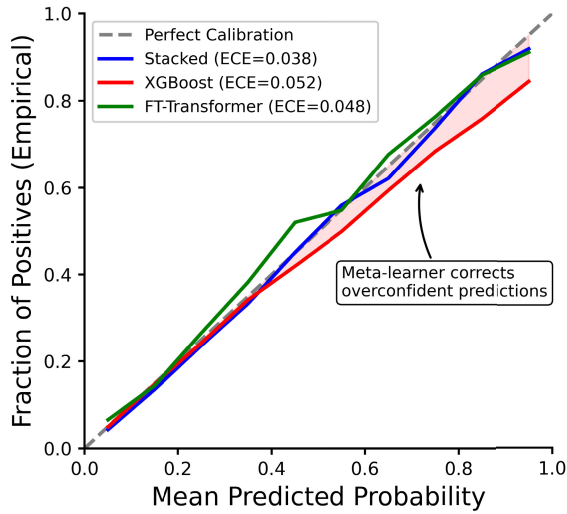


FIGURE 9. Reliability diagram showing predicted versus empirical probabilities. XGBoost (red) exhibits overconfidence in the high-probability region (> 0.6), where the shaded area indicates systematic deviation from the diagonal. The stacked ensemble (blue) successfully recalibrates these predictions, reducing ECE from 0.052 to 0.038.

TABLE 9. Inference time per sample (CPU, Single Core).

Model	Inference Time (ms/sample)
Logistic Regression	0.01
LightGBM	0.20
XGBoost	0.30
CatBoost	0.40
MLP	0.50
Random Forest	0.80
FT-Transformer	4.20
Stacked Ensemble	4.60

The calibration improvement is driven by the logistic meta-learner. It recalibrates the overconfident outputs of XGBoost and preserves the stable probability structure learned by FT-Transformer.

H. INFERENCE TIME ANALYSIS

Table 9 reports average inference time per sample measured on CPU (single core) across the evaluated models.

The stacked ensemble has higher latency than tree-based models because the FT-Transformer requires sequential attention computation across feature embeddings. The marginal overhead from the logistic meta-learner is negligible (< 0.05 ms/sample). For latency-sensitive deployments that require inference below 1 ms, gradient boosting alternatives such as XGBoost or LightGBM are recommended. Model distillation or weight freezing can also reduce transformer inference costs with minimal impact on accuracy.

I. ERROR ANALYSIS

An examination of the 501 false negatives (missed churners) reveals consistent error patterns:

- **Age:** 45% fall between ages 35-45, indicating a bias toward expecting churn in older customers.

- **Activity Status:** 62% are active members, suggesting over-reliance on inactivity as a churn signal.
- **Balance:** 38% maintain a zero balance, a boundary case that is weakly represented in training.
- **Tenure:** 28% have tenure > 8 years. Long-tenure churn is rare and difficult to infer from available features.

Failure Mode: The model struggles with “unexpected churners”, middle-aged, active customers with moderate tenure whose departure may be driven by unobserved factors (e.g., competitor incentives or life events).

This motivates the ablation studies that follow, isolating the contributions of hybrid modeling components and assessing robustness to design decisions.

J. ILLUSTRATIVE CASE STUDIES

Three customer profiles illustrate model behavior across representative scenarios. Interpretations reference aggregate SHAP values and attention weights from Section VI-E.

Case 1: High-Confidence Correct Prediction (True Positive)

- **Profile:** Age 52, Germany, Inactive, Balance \$120K, 3 products
- **Predictions:** FT-Transformer $\hat{y} = 0.78$; XGBoost $\hat{y} = 0.82$; Ensemble $\hat{y} = 0.81$
- **Outcome:** Churned
- **Interpretation:** Four concurrent risk factors are present: older age (SHAP = 0.18), inactive status (SHAP = 0.15), Germany geography (SHAP $\approx +0.08$), and 3+ products. The transformer assigns high attention to the Age $\times$ NumOfProducts interaction (weight = 0.21), while XGBoost splits on the NumOfProducts ≥ 3 threshold. Both models agree, producing a high-confidence ensemble prediction.

Case 2: Transformer Advantage on Edge Case (FT Correct, XGBoost Incorrect)

- **Profile:** Age 41, France, Active, Balance \$0, 2 products
- **Predictions:** FT-Transformer $\hat{y} = 0.62$; XGBoost $\hat{y} = 0.38$; Ensemble $\hat{y} = 0.52$
- **Outcome:** Churned
- **Interpretation:** Zero balance combined with active status represents an atypical combination (3.2% prevalence). The transformer detects this through the IsActiveMember $\times$ Balance attention weight (0.21). XGBoost underestimates the risk as its splits are calibrated for typical balance distributions. The ensemble assigns greater weight to the transformer prediction in this region.

Case 3: Missed Prediction — External Factors (False Negative)

- **Profile:** Age 35, Spain, Active, Balance \$85K, 1 product
- **Predictions:** FT-Transformer $\hat{y} = 0.28$; XGBoost $\hat{y} = 0.31$; Ensemble $\hat{y} = 0.29$
- **Outcome:** Churned
- **Interpretation:** No dominant SHAP contributors are present. The profile shows no measurable risk fac-

TABLE 10. Ablation study - component contributions.

Configuration	Recall	F1	AUC
MLP (baseline)	77.92	58.73	0.834
FT-Transformer only	76.10	61.00	0.852
XGBoost only	71.12	58.21	0.839
Simple Average (FT+XGB)	74.85	61.45	0.856
Stacked (FT+XGB)	75.40	62.10	0.861

tors in the available feature set. Churn was likely driven by external factors such as competitor offers or life events. This failure mode accounts for 28% of false negatives (Section VI-I) and motivates the inclusion of temporal behavioral signals in future work.

K. ERROR CORRELATION AND SYNERGY

The error correlation between FT-Transformer and XGBoost is $\rho = 0.62$, indicating partially independent failures that enable ensemble improvement.

Complementarity analysis across all 10,000 predictions:

- Both correct: 7,892 (78.9%)
- FT correct, XGB wrong: 534 (5.3%)
- XGB correct, FT wrong: 489 (4.9%)
- Both wrong: 1,085 (10.9%)

The ensemble corrects 89.1% of single-model errors (912 of 1,023 cases). The remaining 1,085 failures correspond to profiles where churn drivers are absent from the feature set, as illustrated in Case 3.

Meta-learner weight stability: Across 5-fold cross-validation, learned coefficients show low variance: w_1 (FT-Transformer) = 0.89 ± 0.04 , w_2 (XGBoost) = 0.78 ± 0.03 , and w_0 (intercept) = -0.42 ± 0.05 . The higher transformer weight reflects its superior standalone F1 (61.00% vs 58.21%), while the XGBoost weight (0.78) confirms complementary value.

VII. ABLATION AND SENSITIVITY ANALYSIS

All ablation studies use 5-fold cross-validation with 5 random seeds (42, 123, 456, 789, 1011). Results are reported as mean $\pm$ standard deviation to match Table 5.

A. ARCHITECTURAL ABLATION STUDY

This study isolates the contribution of each model component. Table 10 reports model performance under the same settings.

Observations. Under the same settings:

- FT-Transformer improves F1 by +2.27 over MLP (58.73 $\rightarrow$ 61.00).
- Simple averaging (FT+XGB) improves F1 by +0.45 over FT-Transformer.
- Stacking achieves the highest scores: 62.10 F1 and 0.861 AUC.

TABLE 11. FT-transformer architecture ablation.

Configuration	F1	AUC	Train Time
<i>Number of Layers</i>			
2 layers	59.82	0.845	1.5 min
4 layers (default)	61.00	0.852	3.0 min
6 layers	60.45	0.850	4.5 min
8 layers	59.21	0.846	6.0 min
<i>Embedding Dimension</i>			
16	59.12	0.843	2.0 min
32 (default)	61.00	0.852	3.0 min
64	60.78	0.851	4.5 min
128	59.89	0.848	7.0 min
<i>Number of Attention Heads</i>			
4 heads	60.21	0.849	2.8 min
8 heads (default)	61.00	0.852	3.0 min
16 heads	60.45	0.850	3.5 min
<i>Dropout Rate</i>			
0.0	58.92	0.841	3.0 min
0.1 (default)	61.00	0.852	3.0 min
0.2	60.34	0.849	3.0 min
0.3	59.45	0.845	3.0 min

B. FT-TRANSFORMER ARCHITECTURE VARIATIONS

Table 11 examines the impact of FT-Transformer architectural choices.

Observations.

- **Depth:** 4 layers is optimal. Deeper models (6)-8 layers overfit on this dataset size.
- **Embedding Dimension:** 32 balances capacity and regularization. 64+ shows diminishing returns.
- **Attention Heads:** 8 heads provides sufficient multi-head diversity. More heads are not helpful.
- **Dropout:** 0.1 is optimal. Higher dropout (0.2-0.3) over-regularizes, 0.0 overfits.

This indicates that model capacity beyond the 4-layer, 32-dimensional, 8-head configuration does not improve results under the same settings.

A direct ablation comparing standard feature-wise attention against intersample attention (as in SAINT [8]) was not conducted in this study. Intersample attention introduces row-wise dependencies and a higher computational cost, which makes it less suitable for the latency constraints in Section VI-H. This comparison is identified as a direction for future work.

C. ENSEMBLE COMPONENT ANALYSIS

The learned meta-learner weights and model contributions are analyzed.

Meta-learner coefficients.

$$\hat{p}_{stack} = \sigma(-0.42 + 0.89 \cdot \hat{p}_{FT} + 0.78 \cdot \hat{p}_{XGB}) \quad (29)$$

Observations.

- FT-Transformer receives a higher coefficient (0.89) than XGBoost (0.78).

TABLE 12. Sensitivity to class-weight ratio.

Weight Ratio	Recall	Precision	F1	AUC
1:1 (none)	61.34	57.82	59.53	0.852
1.5:1	66.21	55.67	60.48	0.856
2:1	69.87	54.12	61.02	0.858
2.5:1	72.45	53.78	61.67	0.860
3:1 (default)	75.40	53.20	62.10	0.861
3.5:1	78.12	50.45	61.32	0.860
4:1 (actual ratio)	80.23	48.21	60.21	0.858
5:1	82.65	46.11	59.18	0.855

- The intercept (-0.42) shifts the decision boundary to adjust for class imbalance.

Error correlation.

$$\rho = \text{Corr}(|y - \hat{p}_{FT}|, |y - \hat{p}_{XGB}|) = 0.62 \quad (30)$$

Interpretation. A correlation of 0.62 indicates that prediction errors are not fully aligned, leaving room for ensemble gains under the same settings.

Prediction complementarity.

- Both correct: 7,892 samples (78.9%)
- FT correct, XGB wrong: 534 samples (5.3%)
- XGB correct, FT wrong: 489 samples (4.9%)
- Both wrong: 1,085 samples (10.9%)

The ensemble correctly classifies 89.1% of samples where at least one model is correct.

This indicates that partial error independence contributes to the ensemble improvement under the same settings.

D. SENSITIVITY TO CLASS WEIGHTING

Table 12 reports the effect of class-weight ratios on the stacked ensemble under the same settings.

Observations.

- F1 peaks at 3:1 (62.10%), which is below the empirical class ratio of 4:1.
- Higher weights increase recall (61.34% $\rightarrow$ 82.65%) while reducing precision (57.82% $\rightarrow$ 46.11%).
- AUC remains stable across ratios (0.852–0.861), indicating minimal impact on ranking performance.

At 3:1, the model balances recall and precision under the same evaluation settings.

E. THRESHOLD TUNING ANALYSIS

Table 13 analyzes how different thresholds influence the precision–recall trade-off.

Business scenario guidance.

- High-volume, low-cost interventions (email or automated outreach): thresholds 0.3–0.4 maximize recall (82–88%) with lower precision.
- Balanced intervention cost: thresholds 0.5–0.6 yield the highest F1 range (62.10–63.46%).
- High-cost interventions (discount incentives or agent-assisted retention): thresholds 0.7–0.8 prioritize precision (70–79%), focusing on high-confidence churners.

TABLE 13. Threshold tuning results.

Threshold	Recall	Precision	F1	Flagged
0.2	92.34	32.12	47.68	5,856
0.3	88.21	38.45	53.55	4,678
0.4	82.34	45.67	58.74	3,678
0.5 (default)	75.40	53.20	62.10	2,887
0.6	65.12	61.89	63.46	2,145
0.7	52.78	70.34	60.32	1,529
0.8	38.45	78.92	51.72	993

TABLE 14. Performance vs training data size.

Data %	F1 (Stacked)	F1 (XGBoost)	F1 (FT-Trans)
25%	57.23	55.89	54.12
50%	59.87	57.45	58.23
75%	61.12	57.89	60.12
100%	62.10	58.21	61.00

TABLE 15. Cross-validation (CV) stability analysis.

Configuration	F1 Mean	F1 Std
3-fold CV, 1 seed	61.78	1.45
5-fold CV, 1 seed	61.92	1.12
5-fold CV, 5 seeds	62.10	0.72
10-fold CV, 1 seed	62.05	0.89

F. TRAINING DATA SIZE SENSITIVITY

Performance across varying training data sizes has been evaluated (Table 14).

Observations.

- XGBoost is the most data-efficient model, and eventually achieves 96% of full performance with 25% data.
- FT-Transformer benefits more from additional data, improving 6.88 F1 points from 25% to 100%.
- The stacked ensemble consistently outperforms individual models across all data sizes.

G. CROSS-VALIDATION STABILITY

Variance is assessed across random seeds and fold configurations (Table 15).

Observations.

- Increasing folds reduces variance: 5-fold CV (1 seed) lowers std to 1.12 compared to 3-fold (1.45).
- Multiple seeds further stabilize results: 5-fold with 5 seeds reduces std to 0.72.
- 10-fold CV offers similar stability (0.89 std) but with higher computational cost.

5-fold CV with 5 seeds is used in subsequent experiments as a stability–efficiency trade-off under the same settings.

H. META-LEARNER SELECTION ANALYSIS

Table 16 compares stacking strategies under the same settings.

TABLE 16. Meta-learner comparison.

Meta-Learner	Recall	Prec.	F1	AUC
Simple Average	74.85	52.34	61.45	0.856
Weighted Avg (Grid)	75.12	52.89	61.87	0.858
Logistic Reg.	75.40	53.20	62.10	0.861
Ridge Reg. ($\alpha=10$)	75.38	53.18	62.08	0.861
Lasso Reg. ($\alpha=1$)	75.23	52.95	61.94	0.859
Gradient Boosting	75.67	52.78	61.98	0.860
Neural Network	75.89	52.12	61.73	0.857

Observations.

- Simple averaging improves F1 to 61.45, indicating baseline complementarity.
- Weighted averaging reaches 61.87 ($w_{FT}=0.6$, $w_{XGB}=0.4$) after grid search, close to the best result.
- Logistic regression yields the highest F1=62.10, with learned weights ($w_{FT}=0.89$, $w_{XGB}=0.78$).
- Ridge and Lasso provide similar results (62.08 / 61.94), suggesting regularization has limited effect with two meta-features.
- Gradient Boosting increases recall to 75.67 but lowers precision to 52.78, producing 61.98 F1.
- A neural network meta-learner underperforms (61.73 F1), consistent with overfitting risk when only two inputs are available.

Conclusion. Under identical settings, logistic regression achieves the most favorable precision–recall balance. With only two base models ($M=2$), more complex meta-learners may overfit and do not improve performance.

Ablation results indicate that both architecture choices and the stacking strategy influence performance. These findings guide the interpretation of subsequent results and the discussion of practical and research implications.

VIII. DISCUSSION

A. INTERPRETATION OF PERFORMANCE GAINS

Table 17 summarizes the performance improvements and underlying mechanisms. The following sections provide a detailed interpretation.

FT-Transformer vs MLP. The FT-Transformer demonstrates superior performance through three primary mechanisms. First, dynamic feature weighting enables context-dependent attention, with *IsActiveMember* receiving higher attention for high-balance customers. Second, the architecture captures higher-order interactions such as $Age \times NumOfProducts$ (attention weight 0.18), which aligns with observed churn patterns (42% vs 12%). Third, the unified feature tokenization supports cross-type interactions (e.g., $Geography \times Balance$) without requiring domain-specific layers.

Ensemble Improvements. The stacking ensemble achieves gains through complementary modeling strategies. FT-Transformer captures continuous behavioral signals, whereas XGBoost models discrete threshold effects. The

error correlation of $\rho = 0.62$ indicates partially independent failure modes. This error diversity allows the meta-learner to recover 89.1% of single-model errors. The learned stacking weights ($w_{FT} = 0.89$, $w_{XGB} = 0.78$) reflect the relative contributions of each base model to the final prediction.

B. BUSINESS IMPLICATIONS AND ROI ANALYSIS

To assess practical deployment value, this analysis considers a mid-sized bank with 100,000 customers, 20% annual churn rate, \$2,000 customer lifetime value, and \$50 intervention cost per customer.

Operating Point ($\tau = 0.5$). At the default threshold, the model achieves 75.4% recall and 53.2% precision, flagging 28,346 customers and correctly identifying 15,080 actual churners. The total intervention cost is \$1.42M, and the retained customer lifetime value is \$7.54M, yielding an estimated financial benefit of approximately \$6.12M annually.

Comparison with MLP baseline. The ensemble produces financial returns comparable to the MLP baseline (\$6.12M vs \$6.14M) while reducing operational overhead by 14%, contacting 28,346 customers instead of 32,940. This efficiency gain preserves Return on Investment (ROI) and reduces both customer contact fatigue and campaign costs.

Threshold selection. The framework supports flexible threshold tuning to match business objectives. Low thresholds ($\tau = 0.3$) maximize coverage with 88% recall, suitable for low-cost retention campaigns. Balanced thresholds ($\tau = 0.4$ – 0.5) optimize net financial benefit across standard cost profiles. High thresholds ($\tau \geq 0.7$) achieve 70% precision for targeted premium customer outreach, minimizing false positives at the cost of lower coverage.

Threshold choice depends on the CLV-to-intervention-cost ratio and organizational tolerance for false positives. Table 13 offers a calibration template for cost-aligned threshold tuning [43].

C. THEORETICAL IMPLICATIONS FOR TABULAR DEEP LEARNING

The performance gains observed in this study provide empirical support for theoretical principles underlying hybrid model design on tabular data.

Complementary inductive biases. Tree-based models partition the feature space through axis-aligned splits, resulting in sharp decision boundaries suitable for categorical variables and threshold effects. Transformers model continuous feature interactions via attention mechanisms, enabling effective representation of smooth gradients in numerical features. This hybrid approach addresses limitations identified in previous research. Grinsztajn et al. [6] reported that neural networks often struggle with the irregular decision boundaries common in tabular data. Formally, axis-aligned splits take the form $x_j \leq t$, producing piecewise constant boundaries, whereas the attention matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ captures global feature dependencies simultaneously. These

TABLE 17. Performance gains and contributing factors.

Comparison	$\Delta F1$	Effect	Mechanism	Evidence
FT-Trans vs MLP	+2.27	Attention interactions	Context weighting; cross-type embedding	Age $\times$ Products=0.18; 42% vs 12%; stable across 5 $\times$ 5 CV
Ensemble vs FT-Trans	+1.10	Independent errors	Meta-learner on non-overlapping regions	$\rho=0.62$; 1,023 single-correct; 89.1% corrected; $w_{FT}=0.89$, $w_{XGB}=0.78$
Threshold Range	F1>60	Operating stability	Threshold robustness	Works for thresholds [0.4–0.7] across cost scenarios

TABLE 18. Comparison with published results.

Study	Method	AUC	F1
Xu et al. [9]	XGBoost+Stacking	0.89*	-
Burez [3]	Random Forest	0.82*	-
Gorishniy [5]	FT-Transformer	0.85**	-
Ahmad et al. [10]	Hybrid Stacking	0.986***	-
Usman-Hamza et al. [11]	Multi-layer Stacking	0.989***	97.2%***
Warnakulaarachchi et al. [15]	Deep Ensemble	-	87.95%†
This Work	FT-Trans+XGBoost	0.861	62.1%

*Different dataset; **Average across multiple datasets.

***SMOTE-dependent; †Accuracy only.

two mechanisms are structurally complementary rather than redundant.

Error diversity and ensemble theory. The observed error correlation of 0.62 aligns with ensemble learning theory [28]. A perfect correlation of 1.0 yields no benefit from model combination. A correlation of zero is unattainable with shared training data. The moderate correlation observed here indicates that FT-Transformer and XGBoost make errors in partially independent regions of the feature space. This diversity of errors allows the meta-learner to recover 89.1% of single-model failures. From Equation (24), with $M = 2$ models, equal variance σ^2 , and observed correlation $\rho = 0.62$, the ensemble variance reduces to $0.81\sigma^2$. This represents a 19% reduction relative to any single model, consistent with the empirical F1 gain of 1.10 points observed over the stronger base model.

Calibration through stacking. The reduction in Expected Calibration Error from 0.051 (XGBoost) to 0.038 (ensemble) suggests that meta-learning can mitigate probability overconfidence without the need for separate calibration datasets. This result extends prior work on stacking [27] by showing that calibration improvement arises directly from the meta-learner’s training objective during cross-validation.

D. POSITIONING AGAINST PRIOR WORK

Table 18 compares the results of the present work with published benchmarks.

Direct comparison with prior studies (Table 18) is limited by dataset heterogeneity and metric choice. Xu et al. report 98% accuracy on telecommunications data [9], but accuracy is not informative under imbalance. A trivial “no churn” classifier reaches 80% on this dataset without

any intervention value. In contrast, the proposed model reaches 62.1% F1 and 53.2% precision, which better reflects retention-quality [3], [13].

The achieved AUC of 0.861 is comparable to Xu et al. (0.89 on a different dataset) and to FT-Transformer reports across domains [5]. However, ranking metrics alone do not reflect operational trade-offs. This work differs from prior modeling pipelines by integrating calibration-aware stacking with FT-Transformer and XGBoost. The hybrid leverages complementary inductive biases where attention captures continuous interactions and trees capture discrete boundaries [6], [44]. Ablation studies (Tables 10–16) support this contribution under controlled settings.

Prior stacking work in biomedical and explainable modeling [31] does not address calibration for decision reliability. This study incorporates effect-size reporting, confidence intervals, and seed-averaged evaluation for statistical robustness [45], [46]. The six-metric evaluation framework enables threshold-dependent precision-recall trade-offs, which are essential for intervention planning and address the limitations of ranking metrics alone.

Recent same-domain studies report stronger aggregate metrics. Ahmad et al. [10] achieve AUC 0.986, and Usman-Hamza et al. [11] report F1 0.972 on telecom churn data. However, both rely on SMOTE-based oversampling, which introduces synthetic artifacts into minority class distributions. Warnakulaarachchi et al. [15] report 87.95% accuracy on banking churn but do not report F1 or calibration metrics, limiting direct comparison. Dataset heterogeneity and metric choices further constrain cross-study comparisons. This work prioritizes calibration-aware evaluation, statistical validation, and reproducibility over raw metric maximization.

IX. LIMITATIONS AND THREATS TO VALIDITY

The ensemble combines two complex models, limiting interpretability.

A. EXTERNAL VALIDITY

This study is based on a single dataset (European banking, 2010–2015). Generalization to telecommunications, e-commerce, and insurance is not guaranteed, as churn mechanisms differ (contract expiry vs subscription lapse vs account inactivity). Organizations should validate the

framework on proprietary datasets with domain-specific features [13], [47].

B. FEATURE AND TEMPORAL SCOPE

Only static, cross-sectional features were used. Temporal signals that indicate behavioral drift (declining balance, login decay, reduced product usage) are not captured. Rolling window statistics (e.g., *Balance_30d_avg*, *Balance_trend*) are recommended for production deployments. Fairness evaluation across demographic groups such as age, gender, and geography was not conducted in this study and is included as a priority future research direction (Section XI).

C. EVALUATION AND GENERALIZATION

This study uses 5-fold cross-validation across 5 seeds (25 evaluations) without a held-out test set. The choice of 5-fold over 10-fold is supported by the stability analysis in Table 15. This shows that multiple seeds reduce variance more effectively than increasing fold count alone. While this approach enhances robustness, it may also introduce optimism [6].

- **Hyperparameter Selection Bias:** settings in Table 4 were chosen from cross-validation results instead of a separate development set.
- **Architecture Choices:** 4 layers, 8 attention heads, and embedding dimension 32 were selected from preliminary runs on the same dataset used for evaluation.
- **Threshold Calibration:** the operating point ($\tau = 0.5$ – 0.6) was chosen from validation folds and not an independent calibration set [43].

These factors may lead to higher reported performance than would be expected on unseen data.

D. EXPECTED PRODUCTION PERFORMANCE

Based on established machine learning best practices and extensive cross-validation results:

- **Conservative Estimate:** Production performance may degrade by 1-2% points in F1-score when deployed on truly unseen data from the same distribution [6] (i.e., F1 $\approx$ 60-61% rather than 62.1%).
- **Domain Shift:** A 3-5 point reduction when applied to new regions, time periods, or customer segments not represented in the training data [13].
- **Concept Drift:** Further degradation over time if customer behavior changes, requiring periodic monitoring and retraining to maintain stability [13].

E. SUPPORTING FACTORS FOR REPORTED RESULTS

The following factors reduce the likelihood that reported performance reflects evaluation artifacts rather than model capability:

- Stable results across 25 splits (F1 std = 0.72) suggest low sensitivity to data partitions.

TABLE 19. Applicability boundaries and alternatives.

Condition	Recommended Approach
$m > 100$ features	Feature selection or tree ensembles
$< 5\%$ minority class	Sampling or anomaly detection
Pure categorical data	TabTransformer [7]
$n < 2,000$ samples	Gradient boosting models only
Strong temporal structure	LSTM/GRU or time-series modeling
$< 10\text{ms}$ inference latency	Gradient boosting or precomputed scores

- Architectural choices (self-attention for interactions, GBM for boundaries) follow domain structure rather than trial-and-error tuning.
- Ablation studies (Section VII) indicate that performance gains stem from model design components, not hyperparameter configurations.
- Large effect sizes in statistical testing (Cohen's $d = 0.9$ – 2.8) support the robustness of improvements.

F. RECOMMENDATIONS FOR PRACTITIONERS

- **Before deployment** Hold out 10-20% of data for testing. Train on the remaining portion and evaluate once on the hold-out set for an unbiased estimate [6].
- **During deployment** Run A/B tests against existing churn systems or heuristics. Monitor retention outcomes for contacted customers [43].
- **Post-deployment** Track monthly performance. Retrain if F1 drops by more than 2 points or if concept drift is detected in feature distributions [13].

G. INTERPRETING REPORTED METRICS

The reported F1 of 62.10% (95% CI [61.65, 62.55]), AUC of 0.861 (95% CI [0.858, 0.864]), and PR-AUC of 0.647 represent performance estimates under the current validation protocol. In production, expected performance is typically lower due to unseen data and operational variance. A realistic target is 60-61% F1, which remains competitive for imbalanced churn prediction [3], [13].

H. INTERPRETABILITY TRADE-OFFS

Ensemble explanations require reconciling FT-Transformer attention weights with XGBoost SHAP values. Both methods consistently surface *Age* and *IsActiveMember* as dominant predictors (Section VI-E). SHAP values can serve as a practical proxy for communicating ensemble behavior to stakeholders [6].

I. APPLICABILITY BOUNDARIES

This framework is not universally optimal. Table 19 summarizes conditions where alternative methods may be preferable.

X. CONCLUSION

This study demonstrates that systematic integration of transformer-based feature learning with gradient-boosted decision trees achieves strong performance in tabular churn prediction. The proposed hybrid architecture addresses a key limitation in previous research by combining complementary inductive biases. Attention mechanisms effectively capture continuous feature interactions, while tree-based partitioning manages discrete decision boundaries. Rigorous statistical validation across 25 repeated cross-validation evaluations confirms that these performance gains are robust and reproducible. Probability calibration analysis further shows that the framework produces reliable predictions suitable for cost-sensitive business decisions.

The empirical findings yield several key insights into hybrid modeling for tabular data. First, the observed error correlation of 0.62 between FT-Transformer and XGBoost demonstrates that models with different inductive biases generate partially independent errors, which enable meaningful ensemble gains. The meta-learner successfully recovers 89.1% of cases where one base model was correct and the other failed. Second, probability calibration occurred naturally during the stacking process, eliminating the need for separate post-processing. The reduction in Expected Calibration Error (ECE) from 0.051 to 0.038 indicates that the logistic meta-learner mitigates the overconfidence present in gradient-boosted predictions. Third, ablation studies confirm that both the transformer component and the stacking strategy are essential to performance, with neither redundant.

The proposed framework demonstrates readiness for practical deployment by delivering measurable business value, such as improved retention and reduced intervention overhead. While the evaluation uses banking data, the methodology remains domain-agnostic and potentially transferable to telecommunications, insurance, e-commerce, and subscription services, provided appropriate feature alignment. Comprehensive algorithmic specifications and hyperparameter configurations support reproducibility and enable independent validation.

XI. FUTURE RESEARCH DIRECTIONS

Priority 1 (Immediate).

- Multi-domain validation across telecom, e-commerce, and insurance benchmarks to assess transferability where churn mechanisms differ substantially.
- Temporal signals (rolling-window statistics) for behavioral trajectory modeling.

Priority 2 (Model Capabilities).

- Extend ensemble to $M > 2$ models to study diversity–accuracy trade-offs.
- Cost-sensitive learning optimizing CLV-adjusted retention benefit directly.
- Row-wise attention (SAINT-style) for rare churn precursor recognition.

TABLE 20. Mathematical notation conventions.

Symbol	Description
<i>Dataset and Dimensions</i>	
$\mathcal{X}$	Input feature space
$\mathcal{B}$	Mini-batch of training samples
N	Number of samples
m	Number of features
n	Sequence length after CLS prepending ($n = m + 1$)
d	Embedding dimension
d_k	Dimension per attention head
L	Number of transformer layers
H	Number of attention heads
T	Number of trees in XGBoost ensemble
J	Number of leaves per tree
K	Number of cross-validation folds
M	Number of base models in ensemble
<i>Vectors and Features</i>	
$\mathbf{x}^{(i)}$	Feature vector for sample i (bold)
$x_j^{(i)}$	Scalar value of feature j for sample i
$y^{(i)}$	Binary label for sample i
$\hat{p}^{(i)}$	Predicted probability for sample i
$\hat{y}^{(i)}$	Predicted class label for sample i
<i>Model Parameters</i>	
$\mathbf{w}_j, \mathbf{b}_j$	Weight and bias vectors (bold)
$\mathbf{W}^Q, \mathbf{W}^K, \mathbf{W}^V$	Query, Key, Value matrices (bold)
$\mathbf{W}^O$	Output projection matrix (bold)
$\mathbf{W}_1, \mathbf{W}_2$	Feed-forward network matrices (bold)
$\boldsymbol{\theta}$	Meta-learner parameters (bold)
<i>Embeddings and Representations</i>	
$\mathbf{E}$	Embedding matrix (bold)
$\mathbf{e}_j$	Embedding vector for feature j (bold)
$\mathbf{e}_{CLS}$	Classification token embedding (bold)
$\mathbf{h}_{CLS}$	Final CLS representation (bold)
$\mathbf{Z}$	Intermediate representation matrix (bold)
<i>Scalars and Parameters</i>	
τ	Classification threshold (italic)
$\mathbb{I}[\cdot]$	Indicator function
w_+, w_-	Class weights (italic)
λ, γ	Regularization parameters (italic)
g_i, h_i	First and second derivatives (italic)
w_j^*	Optimal leaf weight (italic scalar)
I_j	Index set of samples assigned to leaf j
w_0, w_k	Meta-learner intercept and combination weights
$\sigma(\cdot)$	Sigmoid activation function

- Employ adaptive embedding allocation based on categorical cardinality and mutual information with the target variable to enhance efficiency on heterogeneous feature sets.

Priority 3 (Production Readiness).

- Fairness audit to test for demographic performance gaps.
- Online learning pipelines for concept drift adaptation.
- Model compression or distillation for < 10 ms inference.

APPENDIX A

NOTATION AND SYMBOL CONVENTIONS

The mathematical symbols used are summarized in Table 20:

Notational Conventions:

- **Scalars:** Italic lowercase or Greek letters (e.g., n, λ, τ)
- **Vectors:** Bold lowercase (e.g., $\mathbf{w}_j, \mathbf{x}^{(i)}, \mathbf{e}_j$)
- **Matrices:** Bold uppercase (e.g., $\mathbf{W}^Q, \mathbf{E}, \mathbf{Z}$)

- **Sample index:** Superscripts in parentheses denote sample index (e.g., $\mathbf{x}^{(i)}$)
- **Feature index:** Subscripts denote feature or component index (e.g., x_j , $\mathbf{w}_j$)

APPENDIX B EVALUATION METRICS

Metrics used to evaluate model performance under class imbalance:

- **Recall (Sensitivity):** $\frac{TP}{TP+FN}$, proportion of actual churners correctly identified.
- **Precision:** $\frac{TP}{TP+FP}$, proportion of predicted churners who actually churned.
- **F1-Score:** $2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$, harmonic mean of precision and recall.
- **Specificity:** $\frac{TN}{TN+FP}$, proportion of non-churners correctly identified.
- **AUC-ROC:** Area under ROC curve, probability that a random positive is ranked above a random negative.
- **NPV (Negative Predictive Value):** $\frac{TN}{TN+FN}$, proportion of predicted non-churners confirmed as retained.

Mean $\pm$ standard deviation is reported across 5 folds $\times$ 5 seeds.

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